Trong Hung Nguyen

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EXPERIENCE

Process Engineer Intern | Walmart Inc. Apple Valley, CA | *Jun 2026 - Aug 2026*

- Diagnosed 550 lost labor hours per month ($206,632 per year) in outbound shipping using Six Sigma DMAIC and a fishbone diagram; presented findings to cross-functional plant leadership.
- Built automated staffing and capacity planning tool for outbound shipping using BigQuery, SQL, Python, and JavaScript; projecting 426 additional cases per hour across a 30-person headcount.
- Traced start-of-shift labor allocation errors to a data-integrity failure in the outbound projection feed; scoped a forecasting model on historical inbound freight to stabilize the input.
- Automated administrative steps in WRQ Host Tool for the Quality Assurance Department using Python and JavaScript; processing 150+ profiles per day and saving 1-2 hours of manual work per shift, 4 days a week.
- Built live KPI dashboards using Python, JavaScript, and SQL; merging 5-6 Ignition SCADA and BigQuery sources into one view, saving operations managers 3-5 hours of manual status checks per shift.

Robotics Operations Engineer Intern | Kiwi Campus Inc. Los Angeles, CA | *Jan 2026 - May 2026*

- Sustained 93% uptime as the delivery fleet scaled 76% (17 to 30 robots, 127+ weekly deliveries) with no added headcount, using preventive maintenance scheduling in Remy.
- Executed 2D LiDAR retrofits at a 95% first-pass rate (5% rework), validating each integration in Remy before redeployment.

Data Analyst | LMU Career and Professional Development Los Angeles, CA | *May 2025 - Present*

- Generate Excel dashboards on 1,857 student engagements across 872 LinkedIn Learning courses, surfacing completion rates and usage patterns that guide advisor resource recommendations.
- Automate engagement-data extraction and missing-field cleanup in Python, cutting processing time 92% versus manual entry while holding 99.5% data integrity.

Lead Undergraduate Researcher | LMU Human Rehabilitation Lab Los Angeles, CA | *Mar 2025 - Feb 2026*

- Led mechanical assembly of the QS-S25 6-DOF motion simulator, cutting assembly time 29% (12 to 8.5 hours) through standard work documentation and modular mounting design.
- Integrated motion feedback into the simulator loop, improving logged driver response time 43% across multiple sessions.
- Fabricated custom throttle pedal in SolidWorks with haptic sensors, achieving 85% realism score in patient survey.

PROJECTS

Capstone: Autonomous Mobility Assistive Robot for Elders Los Angeles, CA | *Feb 2026 - Present*

- Conducted FMEA (IEC 60812) on robot power and electrical system, flagging 24 failure modes and 14 for corrective action
- Compiled 48-line BOM and $5,000 budget across 5 subsystems; cutting power cost by ~$883 to fit budget
- Drafted 21 specs and scored 27 concepts in weighted decision matrices, surfacing a battery pack with +20% run-time margin

Autonomous Vehicle | *TensorFlow, Python, C++* Los Angeles, CA | *Feb 2025 - Aug 2026*

- Built Raspberry Pi vehicle with Python/C++ GPIO/PWM independent 4-wheel steering at under 10 ms latency.
- Mapped environments with 3D LiDAR (Open3D, ICP) and trained a TensorFlow network on the localization data to predict motor commands, reaching 92% path-following accuracy.

Parking Pal | *C++, YOLOv5, ESP32-CAM, Edge Impulse (FOMO), Python* Los Angeles, CA | *May 2025 - Feb 2026*

- Raised detection F1 from 45% to 81% (offline validation) on an ESP32-CAM parking detection system using 5,000+ images labeled with YOLOv5-assisted bounding boxes.
- Cut false positive rate from 28% to 9% with 3,000+ negative samples and data augmentation; deployed FOMO (MobileNetV2 0.35) in C++ at 28 ms inference latency.

TECHNICAL SKILLS

Process & Methods: Lean Manufacturing, Six Sigma, DMAIC, Root Cause Analysis, Continuous Improvement, Process Improvement, Value Stream Mapping, Kaizen, 5S, Standard Work, Capacity Planning, Process Mapping
Data & Analytics: Google BigQuery, SQL, Ignition SCADA, Power BI, Tableau, Microsoft Excel, KPI Dashboarding, Data Integrity
CAD & Design: SolidWorks, CAD, GD&T
ML & Programming: Python, C++, JavaScript, TensorFlow, YOLOv5, Open3D, HTML/CSS, Linux, git
Embedded & Hardware: ESP32, Raspberry Pi, GPIO/PWM, 2D/3D LiDAR
Languages: Vietnamese, English

EDUCATION

Bachelor of Science in Mechanical Engineering | Loyola Marymount University | Los Angeles, CA ***May 2027***
Concentration: Robotics